

AP Precalculus Unit 2 Wkst C
Graphing Tangent and Cotangent

Name: _____

- a) convert each function to AP (phase shift) form and state the phase shift
b) graph one period of each function (see following pages for graphs)
c) is the function increasing or decreasing over the period?
d) using interval notation, describe the rate of change from the left asymptote to the right asymptote, including the points at which it changes, if any (part d appears with the graphs)

1. $y = 2 \tan(2\pi x - \pi) + 1$

2. $y = -3 \tan\left(\frac{\pi}{2}x + \frac{\pi}{2}\right) - 2$

3. $y = \tan\left(2x - \frac{\pi}{6}\right) + 5$

4. $y = 7 \tan\left(x + \frac{\pi}{12}\right)$

5. $y = -2 \tan(4\pi x + 3\pi) - 4$ (no graph in key)

6. $y = \cot\left(\pi x - \frac{\pi}{2}\right) - 3$ (no graph in key)

7. $y = -5 \cot\left(\frac{\pi}{4}x + \frac{\pi}{2}\right) + 2$

8. $y = 3 \cot\left(\frac{\pi}{2}x + \frac{\pi}{2}\right) - 2$

9. $y = -\cot\left(2x - \frac{\pi}{4}\right) + 4$

10. $y = -\frac{1}{2} \cot(3\pi x - \pi) - 2$

1. a) $y = 2 \tan\left(2\pi\left(x - \frac{1}{2}\right)\right) + 1$
 $c = -\frac{1}{2}$, so $x + (-\frac{1}{2})$
 phase shift is $\frac{1}{2}$ units to the right

c) positive tangent is increasing over the period

2. a) $y = -3 \tan\left(\frac{\pi}{2}(x + 1)\right) - 2$
 $c = 1$, so phase shift is 1 unit to the left

c) negative tangent is decreasing over the period

3. a) $y = \tan\left(2\left(x - \frac{\pi}{12}\right)\right) + 5$
 $c = -\frac{\pi}{12}$, so $x + (-\frac{\pi}{12})$
 phase shift is $\frac{\pi}{12}$ units to the right

c) positive tangent is increasing over the period

4. a) $y = 7 \tan\left(x + \frac{\pi}{12}\right)$ already in AP form
 $c = \frac{\pi}{12}$, so phase shift is $\frac{\pi}{12}$ units to the left

c) positive tangent is increasing over the period

5. a) $y = -2 \tan\left(4\pi\left(x + \frac{3}{4}\right)\right) - 4$
 $c = \frac{3}{4}$, so phase shift is $\frac{3}{4}$ units to the left

c) negative tangent is decreasing over the interval

NOTE: for tangent, the phase shift is NOT equal to the Left Boundary

$$6. a) y = \cot \left(\pi \left(x - \frac{1}{2} \right) \right) - 3 \quad c = -\frac{1}{2}, \text{ so phase shift is } \frac{1}{2} \text{ units to the right}$$

c) positive cotangent is decreasing over the period

$$7. a) y = -5 \cot \left(\frac{\pi}{4} (x + 2) \right) + 2$$

$c = 2$, so phase shift is 2 units to the left

c) negative cotangent is increasing over the period

$$8. a) y = 3 \cot \left(\frac{\pi}{2} (x + 1) \right) - 2 \quad c = 1, \text{ so phase shift is 1 unit to the left}$$

c) positive cotangent is decreasing over the period

$$9. a) y = -\cot \left(2 \left(x - \frac{\pi}{8} \right) \right) + 4 \quad c = -\frac{\pi}{8}, \text{ so phase shift is } \frac{\pi}{8} \text{ units to the right}$$

c) negative cotangent is increasing over the period

$$10. a) y = -\frac{1}{2} \cot \left(3\pi \left(x - \frac{1}{3} \right) \right) \quad c = -\frac{1}{3}, \text{ so phase shift is } \frac{1}{3} \text{ units to the right}$$

c) negative cotangent is increasing over the period

NOTE: because the Left Boundary of the parent cotangent function is 0, the phase shift and the new Left Boundary are the same. (Just like it is for sin, cos, sec, and csc.)

remember: for tangent, period of the parent = π , LB OTP = $-\frac{\pi}{2}$, RB OTP = $\frac{\pi}{2}$

1. $y = 2 \tan(2\pi x - \pi) + 1$

midline: $y = 1$

$a = 2$ right, up

period = $\frac{\pi}{2\pi} = \frac{1}{2}$

LB: $2\pi x - \frac{2\pi}{2} = -\frac{\pi}{2}$

$2\pi x = \frac{\pi}{2}$

$x = \frac{1}{4}$

RB: $2\pi x - \frac{2\pi}{2} = \frac{\pi}{2}$

$2\pi x = \frac{3\pi}{2}$

$x = \frac{3}{4}$

inc: $\frac{1}{2}(\frac{1}{4}) = \frac{1}{8}$

d) $(\frac{1}{4}, \frac{1}{2})$ Rot C decreasing
 $(\frac{1}{2}, \frac{3}{4})$ Rot C increasing

changes at
 $x = \frac{1}{2}$

notice: the center of the period is $\frac{1}{2}$ units to the right of 0, which is the center of the period for the parent tangent. this corresponds to the phase shift found in part (a)

2. $y = -3 \tan(\frac{\pi}{2}x + \frac{\pi}{2}) - 2$

midline: $y = -2$

~~a~~ $a = -3$ right, down

period: $\frac{\pi}{\frac{\pi}{2}} = \frac{\pi}{1}(\frac{2}{\pi}) = 2$

inc: $\frac{2}{4} = \frac{1}{2}$

LB: $\frac{\pi}{2}x + \frac{\pi}{2} = -\frac{\pi}{2}$

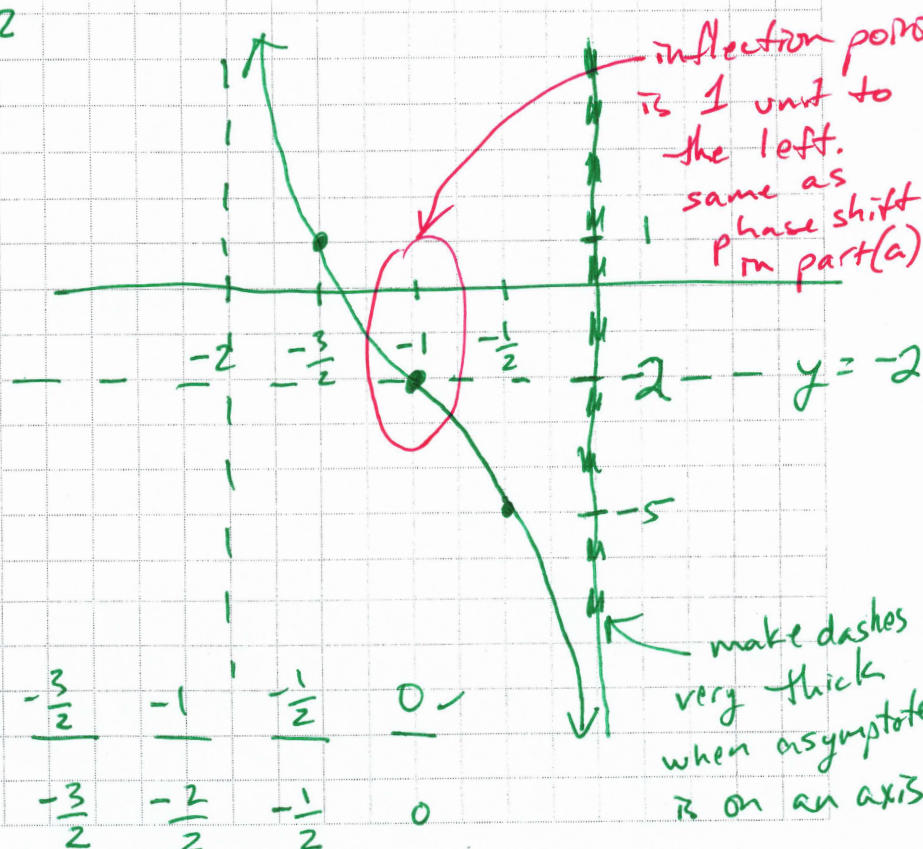
$\frac{\pi}{2}x = -\pi$

$x = -2$

RB: $\frac{\pi}{2}x + \frac{\pi}{2} = \frac{\pi}{2}$

$\frac{\pi}{2}x = 0$

$x = 0$



d) $(-2, -1)$ Rot C ~~decreasing~~ increasing changes at
 $(-1, 0)$ Rot C ~~increasing~~ decreasing $x = -1$

$$3. y = \tan\left(2x - \frac{\pi}{6}\right) + 5$$

midline: $y = 5$

a : 1 right, up

period: $\frac{\pi}{2}$

$$\text{inc: } \frac{\pi}{2} \left(\frac{1}{4}\right) = \frac{\pi}{8} = \frac{3\pi}{24}$$

$$\text{LB: } 2x - \frac{\pi}{6} = -\frac{3\pi}{6}$$

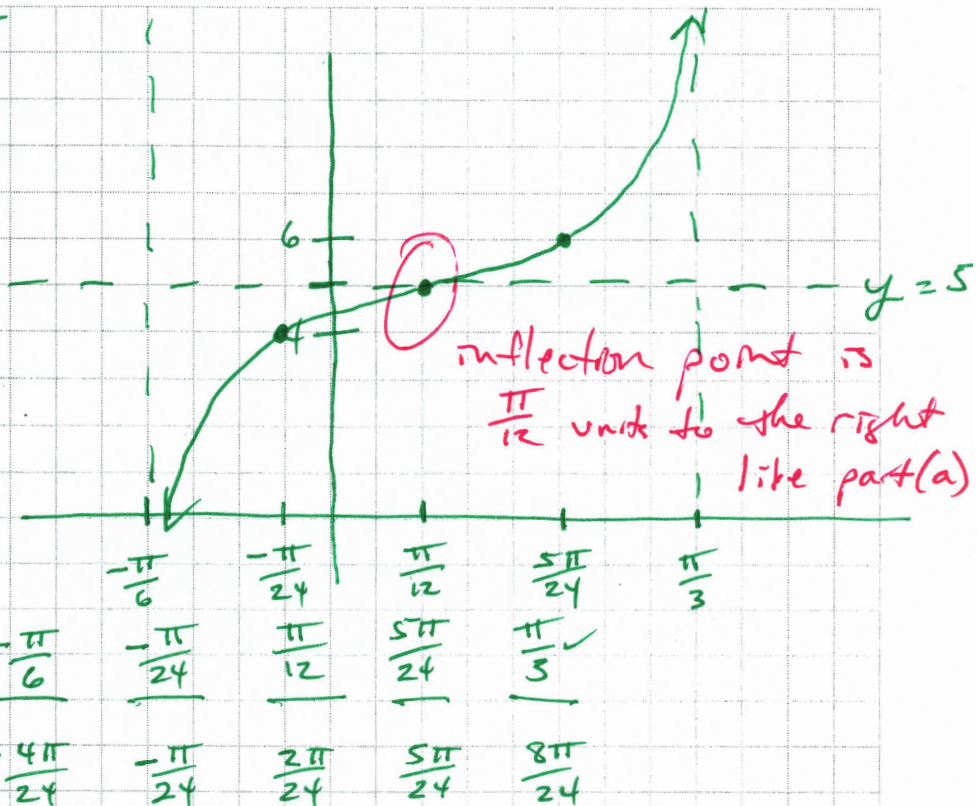
$$2x = -\frac{2\pi}{6}$$

$$x = -\frac{\pi}{6}$$

$$\text{RB: } 2x - \frac{\pi}{6} = \frac{3\pi}{6}$$

$$2x = \frac{4\pi}{6}$$

$$x = \frac{2\pi}{6} = \frac{\pi}{3}$$



(d) $\left(-\frac{\pi}{6}, \frac{\pi}{12}\right)$ Rot C decreasing, $\left(\frac{\pi}{12}, \frac{\pi}{3}\right)$ Rot C increasing
changes at $x = \frac{\pi}{12}$

$$4. y = 7 \tan\left(x + \frac{\pi}{12}\right)$$

midline: $y = 0$

$a = 7$ right, up

period: $\frac{\pi}{1} = \pi$

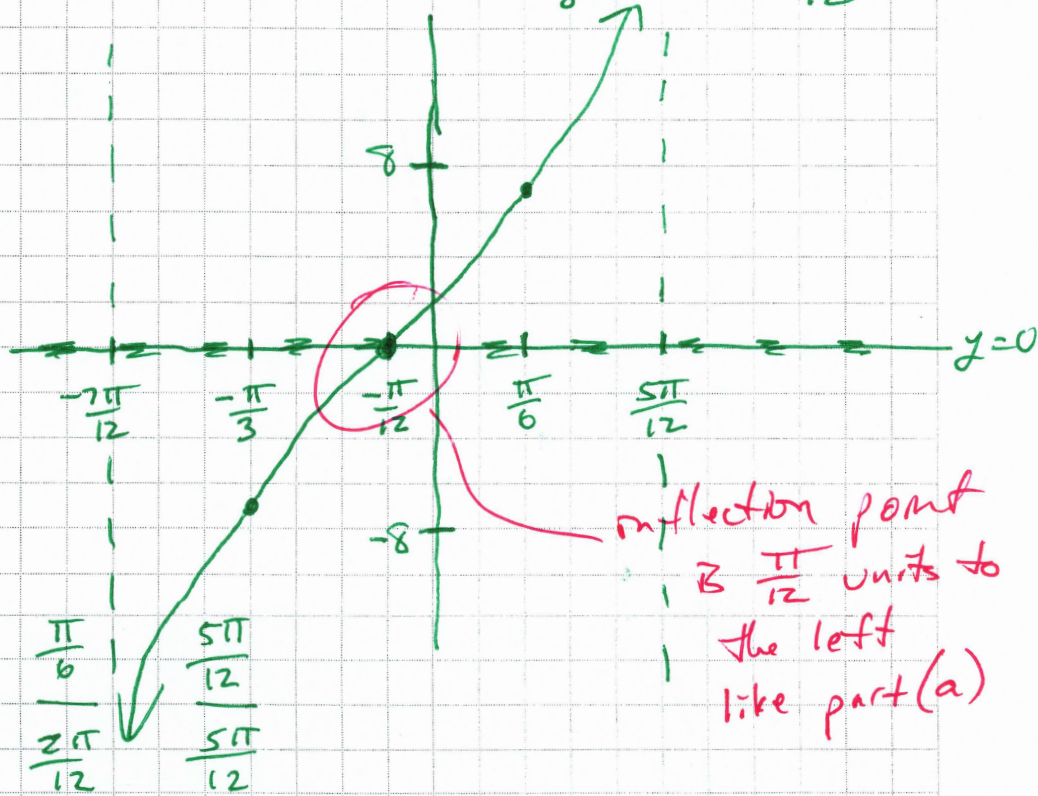
$$\text{inc: } \frac{\pi}{4} = \frac{3\pi}{12}$$

$$\text{LB: } x + \frac{\pi}{12} = -\frac{6\pi}{12}$$

$$x = -\frac{7\pi}{12}$$

$$\text{RB: } x + \frac{\pi}{12} = \frac{6\pi}{12}$$

$$x = \frac{5\pi}{12}$$



(d) $\left(-\frac{7\pi}{12}, -\frac{\pi}{12}\right)$ Rot C decreasing
 $\left(-\frac{\pi}{12}, \frac{5\pi}{12}\right)$ Rot C increasing

changes at $x = -\frac{\pi}{12}$

remember, for cotangent: period of the parent $= \pi$, LB $0\pi = 0$, RB $0\pi = \pi$

7. $y = -5 \cot\left(\frac{\pi}{4}x + \frac{\pi}{2}\right) + 2$

midline: $y = 2$

$a = -5$ left, down

period: $\frac{\pi}{\frac{\pi}{4}} = \frac{\pi}{1} \left(\frac{4}{\pi}\right) = 4$

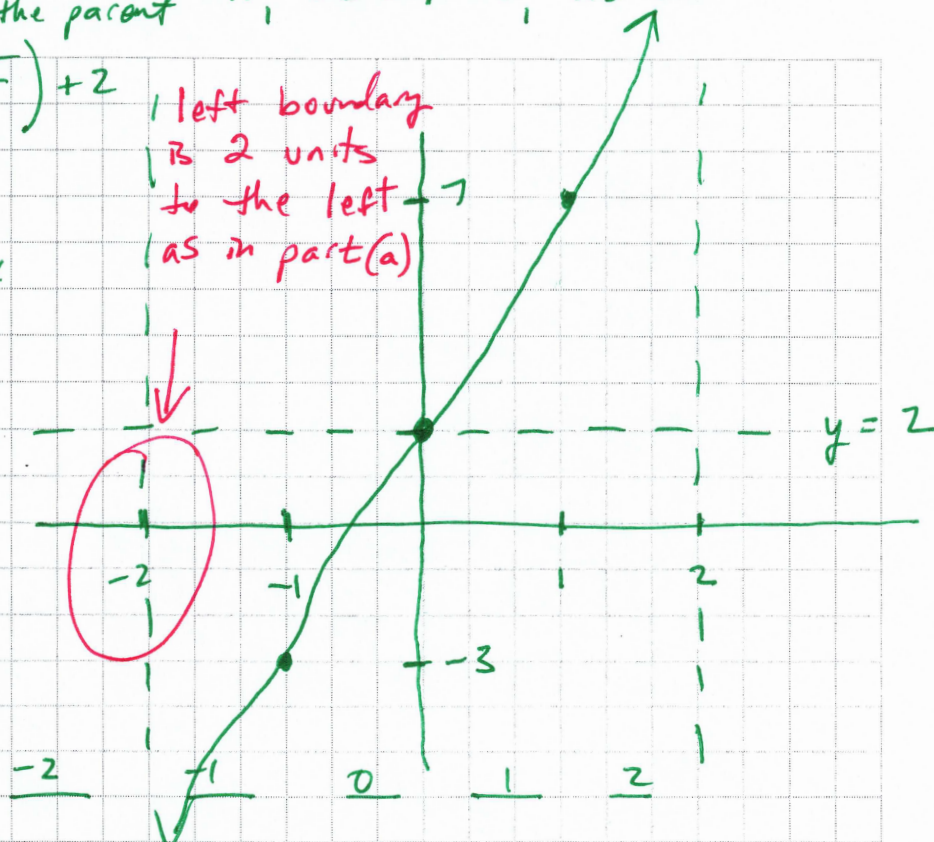
inc: $\frac{4}{4} = 1$

LB: $\frac{\pi}{4}x + \frac{\pi}{2} = 0$

$\left(\frac{4}{\pi}\right)\frac{\pi}{4}x = -\frac{\pi}{2}\left(\frac{4}{\pi}\right)$
 $x = -2$

RB: $\frac{\pi}{4}x + \frac{\pi}{2} = \frac{2\pi}{2}$

$\left(\frac{4}{\pi}\right)\frac{\pi}{4}x = \frac{\pi}{2}\left(\frac{4}{\pi}\right)$
 $x = 2$



(d) $(-2, 0)$ Rot C decreasing
 $(0, 2)$ Rot C increasing

changes at $x = 0$

since cotangent has LB $0\pi = 0$, the phase shift corresponds to the new left boundary, just like for sin, cos, sec, and csc

8. $y = 3 \cot\left(\frac{\pi}{2}x + \frac{\pi}{2}\right) - 2$

midline: $y = -2$

$a = 3$ left, up

period: $\frac{\pi}{\frac{\pi}{2}} = \frac{\pi}{1} \left(\frac{2}{\pi}\right) = 2$

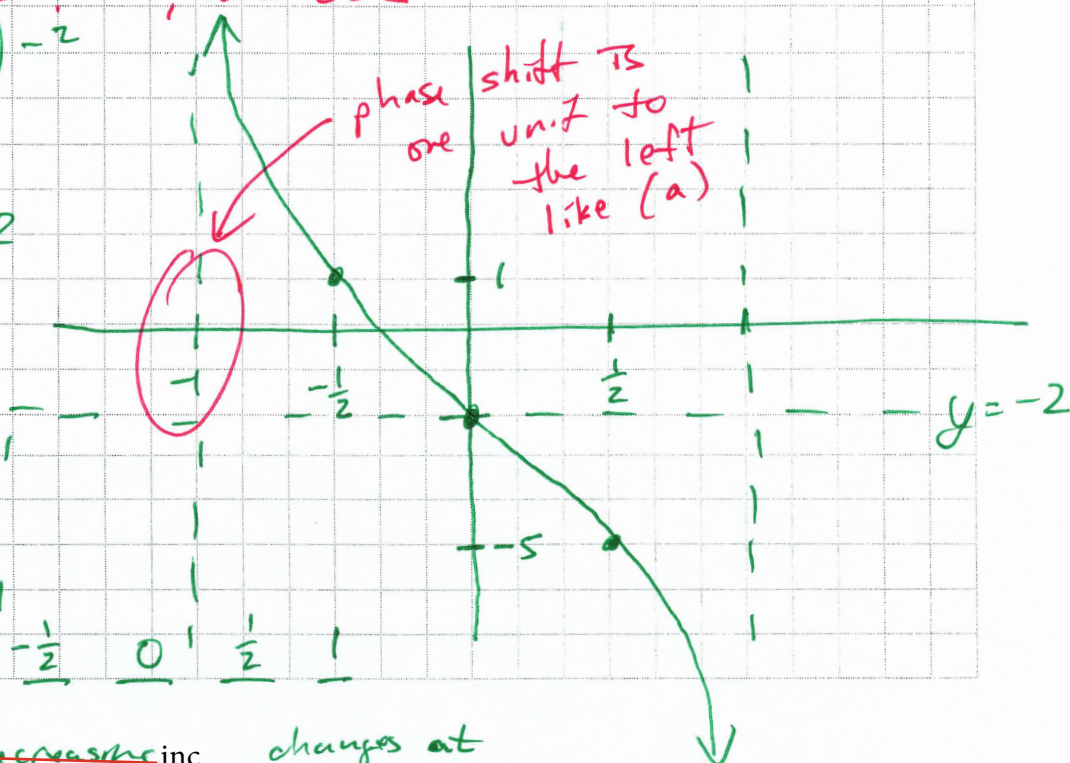
inc: $\frac{2}{4} = \frac{1}{2}$

LB: $\frac{\pi}{2}x + \frac{\pi}{2} = 0$

$\frac{\pi}{2}x = -\frac{\pi}{2}$ $x = -1$

RB: $\frac{\pi}{2}x + \frac{\pi}{2} = \frac{2\pi}{2}$

$\frac{\pi}{2}x = \frac{\pi}{2}$ $x = 1$



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(d) $(-1, 0)$ Rot C ~~decreasing~~ inc changes at
 $(0, -2)$ Rot C ~~increasing~~ dec $x = 0$

$$9. y = -\cot\left(2x - \frac{\pi}{4}\right) + 4$$

midline: $y = 4$

$a: -1$ left, down

period: $\frac{\pi}{2}$

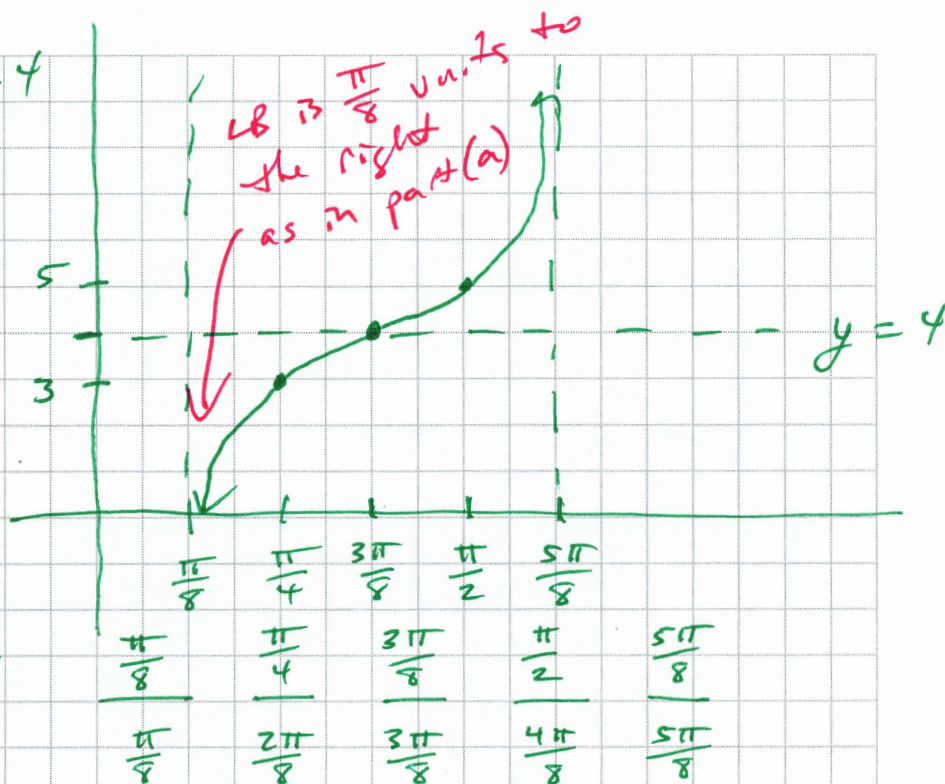
mc: $\frac{\pi}{2}\left(\frac{1}{4}\right) = \frac{\pi}{8}$

LB: $2x - \frac{\pi}{4} = 0$

$2x = \frac{\pi}{4} \quad x = \frac{\pi}{8}$

RB: $2x - \frac{\pi}{4} = \frac{4\pi}{4}$

$2x = \frac{5\pi}{4} \quad x = \frac{5\pi}{8}$



(d) $\left(\frac{\pi}{8}, \frac{3\pi}{8}\right)$ Rot C decreasing
 $\left(\frac{3\pi}{8}, \frac{5\pi}{8}\right)$ Rot C increasing

changes at $x = \frac{3\pi}{8}$

$$10. y = -\frac{1}{2}\cot(3\pi x - \pi) - 2$$

midline: $y = -2$

$a: -\frac{1}{2}$ left, down

period: $\frac{\pi}{3\pi} = \frac{1}{3}$

mc: $\frac{1}{3}\left(\frac{1}{4}\right) = \frac{1}{12}$

LB: $3\pi x - \pi = 0$

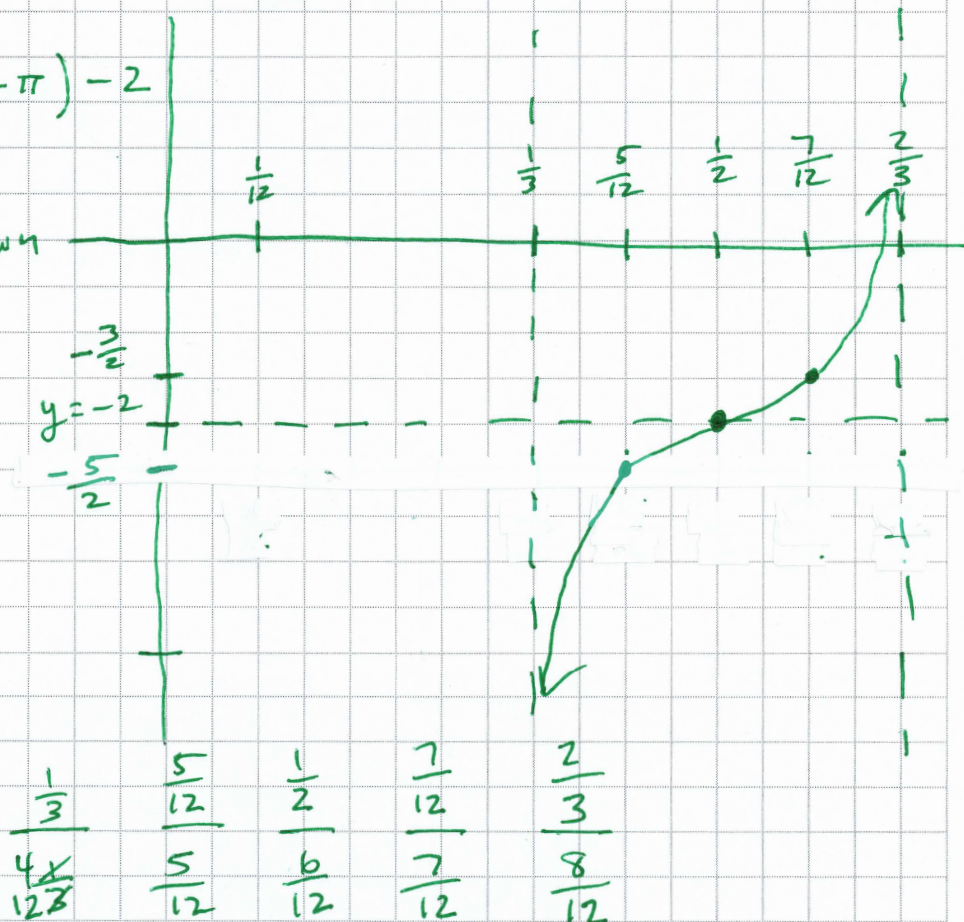
$3\pi x = \pi$

$x = \frac{1}{3}$

RB: $3\pi x - \pi = \pi$

$3\pi x = 2\pi$

$x = \frac{2}{3}$



(d) $\left(\frac{1}{3}, \frac{1}{2}\right)$ Rot C decreasing
 $\left(\frac{1}{2}, \frac{2}{3}\right)$ Rot C increasing

changes at $x = \frac{1}{2}$